RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Thursday, 8 October 2026, 10:11 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:21 AM
Time taken	1 hour 9 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100];
    scanf("%s %s", a, b);

    int n=strlen(a), m=strlen(b), dp[n+1][m+1];

    for(int i=0;i<=n;i++)
        for(int j=0;j<=m;j++)
            dp[i][j]=(i==0||j==0)?0:
            (a[i-1]==b[j-1]?dp[i-1][j-1]+1:
            (dp[i-1][j]>dp[i][j-1]?dp[i-1][j]:dp[i][j-1]));

    printf("%d",dp[n][m]);
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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